

SOC Security Monitoring & Failed Login Detection with Splunk

Security Operations Center (SOC) Project Report

Portfolio Project | Splunk SIEM

Executive Summary

This project demonstrates the use of Splunk as a Security Information and Event Management (SIEM) platform for monitoring authentication activity and investigating suspicious login behavior. The project focused on collecting and analyzing security events, identifying failed and successful login activity, examining source IP addresses and targeted usernames, detecting possible brute-force behavior, configuring security alerts, and presenting findings through a SOC dashboard.

Project Objectives

- Monitor authentication and security-related events in Splunk.
- Identify failed login attempts and successful authentication events.
- Analyze source IP addresses associated with suspicious activity.
- Identify usernames that were repeatedly targeted.
- Investigate patterns that may indicate brute-force attacks.
- Configure security alerts for suspicious authentication activity.
- Create a SOC dashboard to present relevant security findings.
- Demonstrate a practical workflow for SOC monitoring and investigation.

Tools and Technologies

Tool / Technology	Purpose
Splunk	SIEM platform used for log collection, search, analysis, alerting, and dashboarding.
Authentication/Security Logs	Event data used to investigate login activity and identify suspicious patterns.
SOC Investigation Workflow	Process of monitoring, triaging, investigating, and communicating security events.

Project Environment

The project was developed as a hands-on SOC laboratory exercise using Splunk to simulate security monitoring and authentication-event investigation. The analysis centered on login activity and the identification of abnormal authentication patterns.

Investigation and Detection Workflow

Stage	Activity
Log Monitoring	Security and authentication events were reviewed in Splunk to establish visibility into login activity.
Failed Login Analysis	Failed authentication events were examined to identify repeated unsuccessful attempts.
Successful Login Analysis	Successful authentication events were reviewed to provide context around account activity.
Source IP Analysis	Source IP information was examined to identify systems generating suspicious authentication activity.
Username Analysis	Frequently targeted usernames were reviewed to identify possible account-targeting patterns.
Brute-Force Detection	Repeated failed authentication attempts were analyzed as a potential indicator of brute-force behavior.
Alerting	Security alerting was configured to help surface suspicious authentication activity.
Dashboarding	Relevant findings were organized into a SOC dashboard for easier monitoring and investigation.

Key Investigation Areas

- Failed login attempts: Used to identify repeated authentication failures and suspicious activity.
- Successful logins: Used to correlate successful access with preceding authentication failures.
- Top source IPs: Used to identify systems generating the highest volume of authentication attempts.
- Most targeted usernames: Used to identify accounts receiving repeated login attempts.
- Brute-force behavior: Used to identify repeated authentication failures that could indicate credential attacks.
- Security alerts: Used to improve the SOC's ability to identify suspicious events.
- SOC dashboard: Used to present monitoring and investigation results in a consolidated view.

Findings and Security Significance

The project demonstrates how authentication logs can be transformed into actionable security information. Repeated failed login attempts, unusual source activity, and patterns involving specific usernames can provide early indicators of credential attacks. Correlating failed and

successful authentication events can also help a SOC analyst determine whether suspicious activity may have resulted in unauthorized access.

SOC Analyst Response Approach

1. Validate the alert and determine whether the authentication activity is expected.
2. Review the source IP, targeted username, time pattern, and volume of attempts.
3. Correlate failed and successful authentication events.
4. Determine whether the behavior is consistent with brute-force or password-guessing activity.
5. Escalate confirmed suspicious activity for further investigation.
6. Document findings and recommended containment or remediation actions.

Evidence and Screenshots

The GitHub project contains the supporting screenshots for the Splunk lab. The following evidence should be included in the final GitHub report when available:

- Splunk search showing failed login events.
- Splunk search showing successful login events.
- Analysis of top source IP addresses.
- Analysis of most targeted usernames.
- Brute-force detection/search results.
- Configured security alert.
- Final SOC monitoring dashboard.

Note: Insert the corresponding screenshots from the project's screenshots folder under each evidence item. The report intentionally does not invent numerical findings that are not currently documented in the project.

Skills Demonstrated

- Splunk SIEM usage
- Security log analysis
- Authentication event analysis
- Failed-login monitoring
- Brute-force attack detection
- Security alert configuration
- SOC investigation
- Security dashboard development
- Incident triage and analytical thinking

Recommendations

- Monitor repeated failed authentication attempts and establish appropriate alert thresholds.
- Use correlation between failed and successful logins to identify possible account compromise.
- Investigate unusual or high-volume authentication sources.
- Apply account lockout or adaptive authentication controls where appropriate.
- Use strong password policies and multi-factor authentication to reduce credential-based attacks.
- Continue improving dashboard and alert logic to reduce false positives and improve analyst response.

Conclusion

This Splunk SOC project provided practical experience in using SIEM capabilities to monitor authentication events, investigate suspicious login behavior, identify potential brute-force activity, configure alerts, and present security information through a dashboard. The project demonstrates a foundational SOC workflow from event visibility and detection through investigation and reporting.